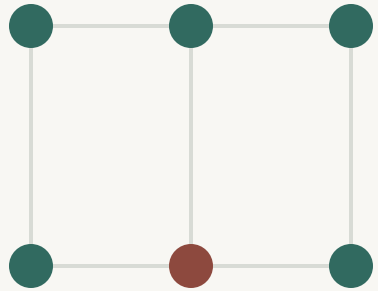
EdgeTrim-KD

Can teacher-guided edge removal make GNNs cheaper while preserving node-classification accuracy?



A proposal for sparse graph learning with knowledge distillation

Sparse graphs promise lower inference cost

how to remove edges without losing the useful message-passing structure.

A GNN aggregates information along edges. Fewer edges can reduce computation, but indiscriminate removal can erase useful neighborhood signals.



Research question

Can a teacher identify which connections matter?

Success criterion

Measure accuracy and inference cost at matched edge budgets.

Prior work already prunes edges using learned and structural rules

Tests teacher-loss scoring plus distillation at matched budgets.

JOINT PRUNING

Joint edge + weight

Learns which edges and parameters to remove together. Chen et al. (2021); Unifews (2025) [1, 4].

STRUCTURAL HEURISTIC

Degree-based

TEDDY uses degree-based discrimination to trim edges. Seo et al. (2024) [2].

LEARNED SPARSIFIERS

Node-adaptive

Mixture-of-graphs sparsifiers adapt edge choices by node. Zhang et al. (2025) [3].

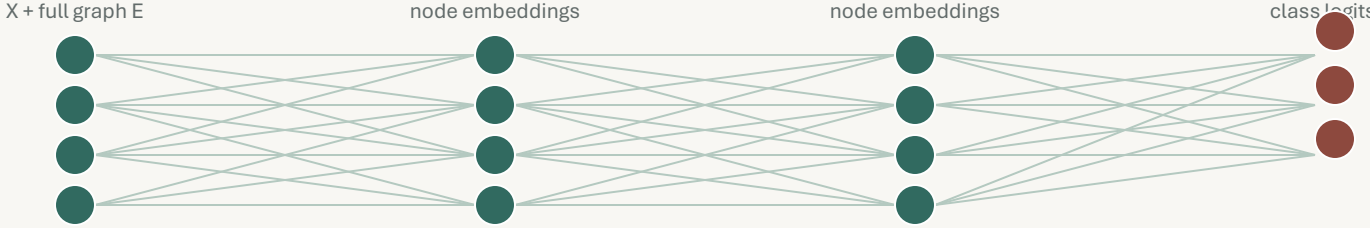
Proposed direction:

Test teacher-loss edge scoring plus distillation at the same edge budget.

The student is a smaller GCN with fewer hidden units

Both process the same node features; the student has fewer trainable parameters and uses the retained graph.

TEACHER / FULL GRAPH



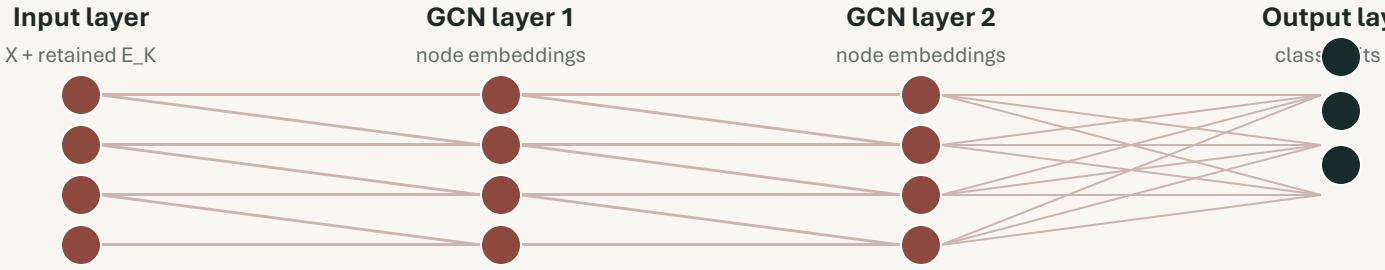
soft targets pT(T)

teacher output

EDGE SCORING / KEEP THE TOP-K CONNECTIONS

Edge gates m_e — one weight per edge → **Teacher loss L_T** — training labels → **Edge score s_e** — |dL_T / dm_e| → **Top-K budget** — keep highest scores → **Sparse graph E_K** — retained connections

STUDENT / SPARSE GRAPH



Student loss

CE(y) + KD(pT, pS)

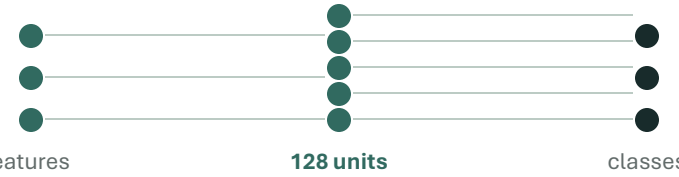
Same sampled graph nodes; model sizes are compared on the next slide.

Teacher scoring and distillation are proposed; current results are supervised pruning baselines.

Teacher 128 → student 64

TEACHER · FULL GRAPH

2-layer GCN · 128 hidden units



Original input + full graph

Teacher produces labels + edge scores

edge scores + soft targets

STUDENT · SPARSE GRAPH

2-layer GCN · 64 hidden units



Same input + retained edges

Labels + teacher targets; fewer parameters

The Cora baselines share one supervised training recipe

The planned distillation run adds teacher targets after the baseline protocol is fixed.

CURRENT BASELINES

Model	2-layer GCN · hidden width 128 · dropout 0.5
Optimizer	Adam · learning rate 0.01 · weight decay 0.0005
Selection	Best validation accuracy · patience 50 · up to 300 epochs
Loss	Cross-entropy on training-node labels only

PLANNED EDGETRIM-KD RUN

1 Train / load full-graph teacher

2 Select edges at a fixed budget

3 Train narrower sparse student + KD

$L = (1 - \alpha) * CE + \alpha * T^2 * KL(pT \,||\, pS)$

4 Compare accuracy, latency, memory

Cora verifies the pipeline; ogbn-arxiv tests scale

The three-seed ogbn-arxiv baseline run is complete; teacher-guided pruning and distillation remain future work.

Cora	2,708 nodes / 1,433 features / 7 classes
ogbn-arxiv	169,343 nodes / 128 features / 40 classes
Splits	Cora public split; ogbn-arxiv official year split
Graph prep	ogbn-arxiv made undirected and coalesced; 1,157,799 connections
Run coverage	Full, random, degree; three seeds on NVIDIA A30
Metrics	Test accuracy, training-loop time, peak GPU memory

What remains to test

These are the proposed teacher-guided pruning and distillation experiments.

01	**Score edges**	Rank connections by sensitivity of the teacher loss.
02	**Prune to a fixed budget**	Keep the top-scoring edges and measure the retained graph.
03	**Train a smaller student**	Use 64 hidden units versus the teacher's 128, with labels + soft targets.
04	**Compare fairly**	Test same-width and smaller students; measure accuracy, latency, memory, and scoring cost.

Current results are pruning baselines; the teacher-scoring and distillation stages are next.

Degree pruning retains more Cora accuracy at the same edge count

Preliminary single-seed results · test accuracy and median forward-pass latency

Method	Edge entries kept	Test accuracy	Median inference
Full graph	10,556	81.1%	1.650 ms
Random pruning	6,258	76.1%	1.120 ms
Degree pruning	6,258	**79.3%**	1.618 ms

Random removes about 40.7% of edge entries; degree keeps the same count for a matched-budget comparison.

Latency is noisy in this small run. Treat it as descriptive; do not claim a speedup yet.

Degree pruning stays close to full-graph accuracy at 60% of edges

Three seeds on NVIDIA A30 / mean +/- population standard deviation

Method	Connections	Test accuracy	Train-loop	Peak memory
Unpruned	1,157,799	67.17 +/- 0.04%	19.96 s	2.95 GB
Random (60%)	694,679	65.18 +/- 0.12%	13.21 s	1.97 GB
Degree (60%)	694,679	**66.88 +/- 0.04%**	12.87 s	1.97 GB

At 40% edge removal, degree pruning is 0.30 percentage points below full accuracy; random pruning is 1.99 points below.

Degree pruning cuts training-loop time by 35.5% and peak memory by 33.2% versus unpruned.

These are training metrics. ogbn-arxiv inference latency and edge-scoring cost were not reported.

Preserve label accuracy with fewer edges and a smaller student

Teacher scores edges; the student learns from labels and the teacher's softened predictions.

OBJECTIVE

Keep the useful signal

1. Score edges by teacher-loss sensitivity
2. Keep top-K connections under a fixed budget
3. Train the 64-unit student on the sparse graph

STUDENT TRAINING LOSS

$$L = (1 - \alpha)\, CE(y, pS)$$

$$+ \alpha\, T^2\, KL(pT(T) \,||\, pS(T))$$

CE: fit ground-truth labels

KL: match softened teacher outputs

α balances the two signals; T softens predictions.

What should the student optimize first?

Accuracy and fidelity can disagree, especially when the teacher is uncertain or wrong.

TASK PERFORMANCE

Accuracy

Match ground-truth node labels

Measure: validation / test accuracy

TEACHER FIDELITY

Resemblance

Match the teacher's soft predictions

Measure: KL divergence / agreement

If they conflict, which should be our primary objective?

Any questions?